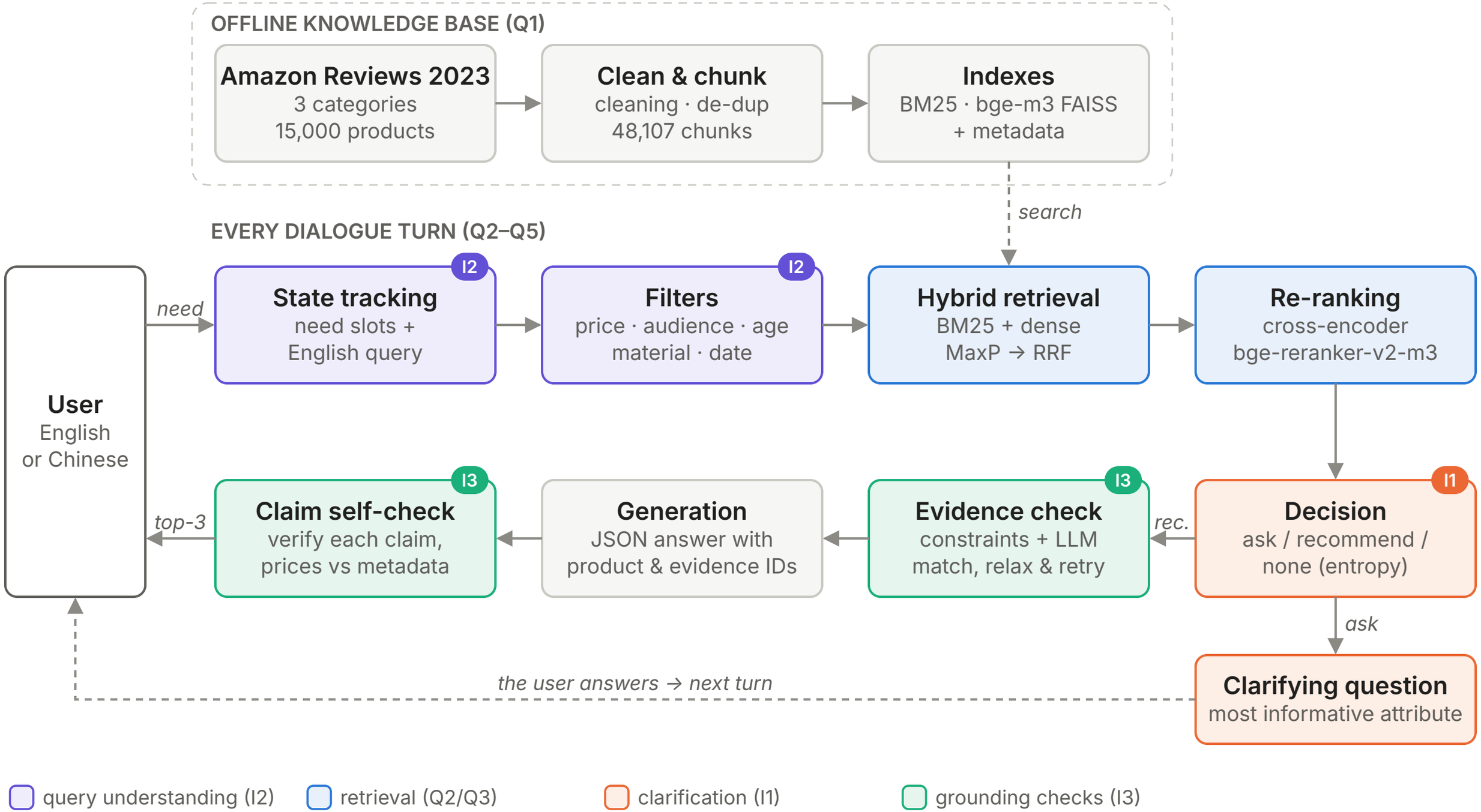
OFFLINE KNOWLEDGE BASE (Q1)
Amazon Reviews 2023
3 categories
15,000 products
Clean & chunk
cleaning · de-dup
48,107 chunks
Indexes
BM25 · bge-m3 FAISS
+ metadata
search
EVERY DIALOGUE TURN (Q2–Q5)
User
English
or Chinese
need
I2
State tracking
need slots +
English query
I2
Filters
price · audience · age
material · date
Hybrid retrieval
BM25 + dense
MaxP → RRF
Re-ranking
cross-encoder
bge-reranker-v2-m3
I1
Decision
ask / recommend /
none (entropy)
rec.
I3
Evidence check
constraints + LLM
match, relax & retry
Generation
JSON answer with
product & evidence IDs
I3
Claim self-check
verify each claim,
prices vs metadata
top-3
ask
Clarifying question
most informative attribute
the user answers → next turn
query understanding (I2)
retrieval (Q2/Q3)
clarification (I1)
grounding checks (I3)